

CIT-EVAL: Development and Implementation of a Secure, Privacy-Preserving Online Faculty Evaluation System with EC-DH and AES Integration

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Abstract. This study presents the design and development of CIT-EVAL, a secure and privacy-preserving online faculty evaluation system for the College of Information Technology (CIT) at the University of the Assumption. Traditional faculty evaluation systems transmit student feedback in plain text, exposing sensitive data to breaches and administrative access, which discourages students from providing honest responses. CIT-EVAL addresses this privacy gap by implementing a Zero-Knowledge cryptographic architecture using the Elliptic-Curve Diffie-Hellman (ECDH) key agreement protocol and AES-GCM 256-bit encryption. Feedback is encrypted directly in the student's browser before transmission, ensuring that even database administrators cannot access raw comments without the appropriate private key. The system was developed using a Waterfall methodology with a descriptive-developmental research design. The backend was built using Java 17 and Spring Boot, while the frontend was developed with React 19 and TypeScript. Access is restricted to verified institutional accounts through Google OAuth 2.0 domain filtering. Pilot deployment within the CIT department demonstrated that the system successfully delivers secure, anonymous, and accessible faculty evaluations while providing administrators and faculty with real-time analytics through a dedicated control center and faculty dashboard.

Keywords: *online faculty evaluation, end-to-end encryption, ECDH, AES-GCM, data privacy, Zero-Knowledge architecture, web application, educational technology*

1 Introduction

1.1 Background of the Study

Traditional faculty evaluation systems in academic institutions often rely on paper-based forms or basic web-based platforms that transmit and store student feedback in plain text. This approach creates a significant privacy vulnerability: any database breach or unauthorized administrative access can expose sensitive student responses. This 'privacy gap' discourages students from providing honest and critical feedback, which ultimately undermines the purpose of the evaluation process.

The CIT-EVAL project was developed in direct response to this challenge. By moving the encryption boundary to the client side, the system ensures that student feedback is encrypted

inside the browser before it is ever sent to the server. This Zero-Knowledge approach means that even the system administrators cannot read raw student comments without the appropriate private key. As academic institutions increasingly recognize the importance of data privacy in their internal processes, developing systems that protect the integrity and confidentiality of evaluation data has become essential.

The system integrates advanced cryptographic protocols, specifically the Elliptic-Curve Diffie-Hellman (ECDH) key agreement protocol and AES-GCM 256-bit authenticated encryption to establish a secure communication channel between the student's browser and the institutional server. This approach provides Perfect Forward Secrecy (PFS), ensuring that future server compromises cannot expose past evaluation data, while maintaining low computational overhead for mobile and web clients.

1.2 Objectives of the Study

This study aims to design, develop, and implement a secure, privacy-preserving online faculty evaluation system specifically for the CIT department at the University of the Assumption. The primary objective is to build a web-based platform capable of handling online form submissions, robust data storage, and real-time analytics while ensuring maximum data integrity through advanced cryptographic standards. To protect the confidentiality of qualitative student feedback, the system will implement client-side encryption using **ECDH (P-256)** and **AES-GCM (256-bit)** prior to transmission. Furthermore, the project involves developing an integrated Admin Control Center for managing faculty accounts and evaluation criteria, alongside personalized dashboards for faculty members to view anonymized feedback and per-criterion performance analytics. Finally, to maintain a secure and exclusive environment, system access will be strictly limited to verified institutional members via **Google OAuth 2.0** domain filtering.

1.3 Conceptual Framework

This study is anchored on the principles of Privacy by Design (PbD), a framework that advocates for privacy to be embedded into the design of systems from the ground up rather than added as an afterthought. CIT-EVAL operationalizes this by implementing client-side encryption as a core architectural decision rather than an optional security layer.

The study also draws from the Software Development Life Cycle (SDLC) using the Waterfall model, which provides a structured and sequential approach to system development. This framework guides the project through clearly defined phases: requirements gathering, system design, development, testing, deployment, and final evaluation.

The overall system flow follows an Input-Process-Output (IPO) model. The input consists of student credentials, evaluation ratings, and written comments. The process involves client-side encryption, secure transmission over HTTPS, server-side storage, and authorized decryption. The output includes encrypted evaluation records, real-time analytics, and faculty performance dashboards accessible to authorized personnel.

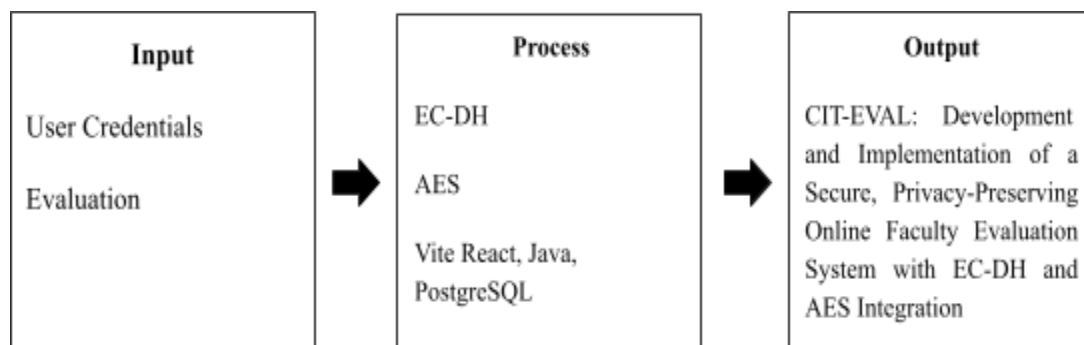


Figure 1. Conceptual Framework

2 Methods

2.1 Research Design and Model Used

This study employs a descriptive-developmental research design. The descriptive component analyzes the limitations of existing evaluation methods specifically, the exposure of plain-text student feedback and the resulting reluctance of students to provide honest responses. The developmental component focuses on the actual construction of the CIT-EVAL system as the solution to these identified problems.

The system was developed using the Waterfall model, a sequential development approach where each phase must be fully completed before the next begins: Requirements Gathering, System Design, Development, Testing, Deployment, and Evaluation. This model was appropriate given that the system requirements, security specifications, and project timeline were clearly defined from the outset.

2.2 System Development Methodology

The Waterfall methodology was implemented through six sequential phases. During the Requirements Gathering phase, the limitations of traditional evaluation systems were analyzed to identify the core features and security specifications the system needed to address. In the System Design phase, the database schema, UI prototypes, and cryptographic architecture were planned, including the ECDH key exchange workflow and AES-GCM encryption pipeline.

The Development phase involved building both the backend REST API using Java 17 and Spring Boot 3.4.4, and the frontend Single Page Application (SPA) using React 19 and TypeScript. PostgreSQL with citext support was used as the primary database. Authentication was handled through Google OAuth 2.0 restricted to the @ua.edu.ph domain, and all encryption operations were implemented using the Web Crypto API.

Following development, the Testing phase verified the functionality of all major system components, including user authentication, the six-stage cryptographic workflow, evaluation form submission, and the Admin Control Center analytics. The system was then deployed within the CIT department for pilot use, followed by a final Evaluation phase where user feedback and system performance data were gathered.

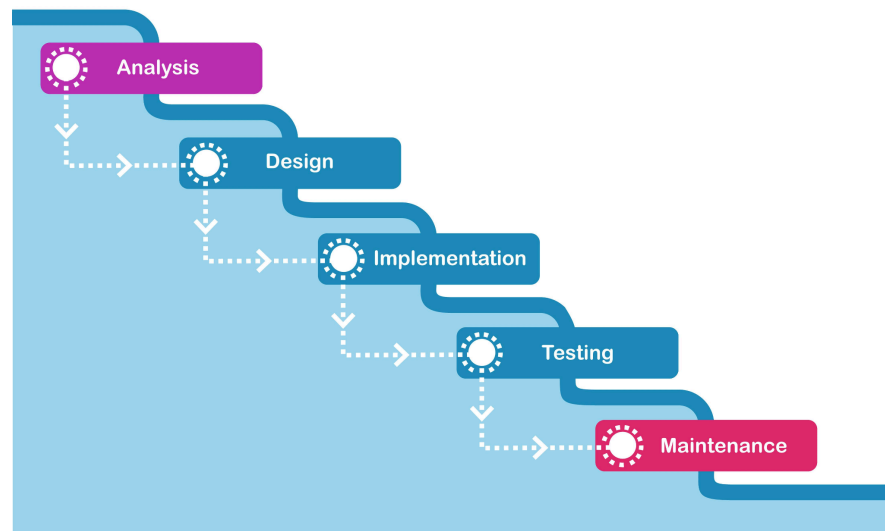


Fig. 2. Waterfall model

Table 1. Technology Stack and Specifications

Layer	Technology
Backend Framework	Java 17, Spring Boot 3.4.4
Frontend Library	React 19, Vite, TypeScript
Primary Database	PostgreSQL (with citext support)
UI Framework	Material UI (MUI), Framer Motion
Authentication	Google OAuth 2.0 (UA Domain Restricted), JWT
Encryption Standard	Web Crypto API: ECDH (P-256) & AES-GCM (256-bit)

2.3 Data Collection Procedure

Data was collected across three key phases. First, a requirements analysis was conducted to examine the limitations of traditional evaluation systems and define the core features the new system needed to address. Second, system testing was performed after development to verify that all features including authentication, encryption, form submission, and analytics functioned correctly and securely. Finally, user feedback was gathered as students, faculty, and administrators interacted with their respective portals to assess the system's usability and performance.

2.4 Ethical Considerations

The study is committed to protecting the rights and privacy of all users. Student feedback is encrypted in the browser before transmission, ensuring that even administrators cannot read raw comments without the server's private key. Access is strictly limited to verified @ua.edu.ph accounts through Google OAuth 2.0 to prevent unauthorized use. All submissions are protected from tampering through AES-GCM authenticated encryption, and the system tracks duplicate submissions without linking a student's identity to the actual content of their feedback. HTTPS is required for all data exchange to ensure secure transmission.

3 Results and Discussion

3.1 System Implementation

The CIT-EVAL system was successfully developed and deployed within the CIT department. The system supports three distinct user roles: students, faculty, and administrators, each with a dedicated interface tailored to their specific functions.

Students access the system by authenticating via their UA Google account. After verifying their identity through the Student Identification Page, they are presented with evaluation forms for their assigned faculty. Numerical ratings are submitted as standard integers, while qualitative comments are processed through the ECDH encryption engine before submission. A review page allows students to confirm their entries before final secure submission.

The cryptographic workflow follows six stages: (1) the frontend retrieves the server's static public EC key; (2) the browser generates a one-time-use ephemeral key pair; (3) the browser derives a unique shared secret; (4) the shared secret is transformed into a 256-bit AES key; (5) the student's comment is encrypted using AES-GCM with a unique Initialization Vector (IV); and (6) the encrypted ciphertext, the student's public key, and the IV are transmitted to the backend.



Fig. 3. Login Page

Evaluation Proper. The Evaluation Proper is the part of the system where students fill out their forms. It is a step-by-step process where students give scores to their teachers, write their comments, and submit them safely using high-level security.

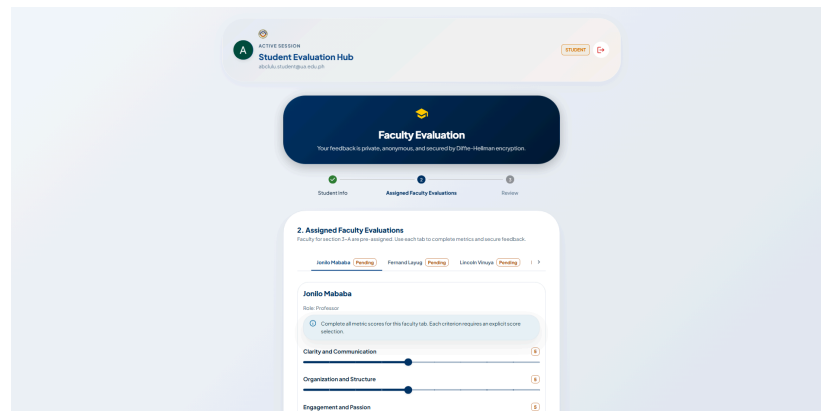


Fig. 4. Evaluation Proper

Admin Control Center. The Admin Control Center provides administrators with a comprehensive dashboard for managing the evaluation process. Administrators can monitor overall evaluation insights, manage faculty accounts and their assigned sections, configure evaluation criteria, and view encrypted feedback submissions. The dashboard displays key metrics including total encrypted submissions, faculty accounts, and active evaluation criteria. Faculty rankings, section trends, and performance comparisons are visualized through charts and graphs to support data-driven decision-making.

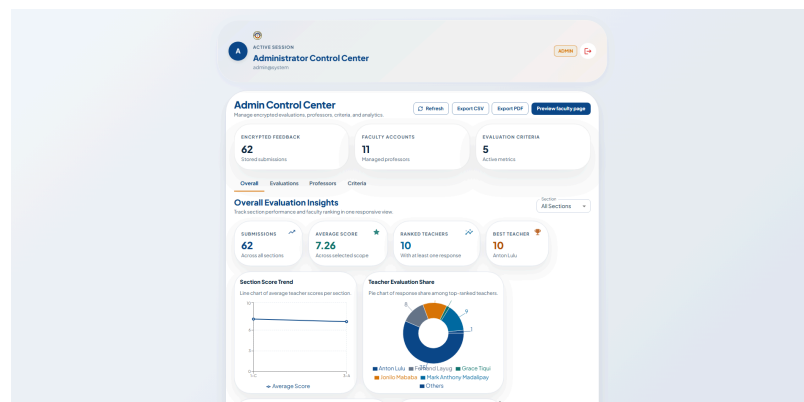


Fig. 3. Admin Dashboard

Faculty Dashboard. Faculty members can access their personal portal to view their performance scores, student participation counts, and per-criterion analytics. The dashboard presents a Criteria Average Radar chart highlighting strengths and areas for improvement, alongside a Section Trends chart comparing average performance and response volume by section. Detailed student feedback is displayed with student identities fully anonymized, showing only the section and the decrypted comment, ensuring honest feedback while maintaining evaluator privacy.

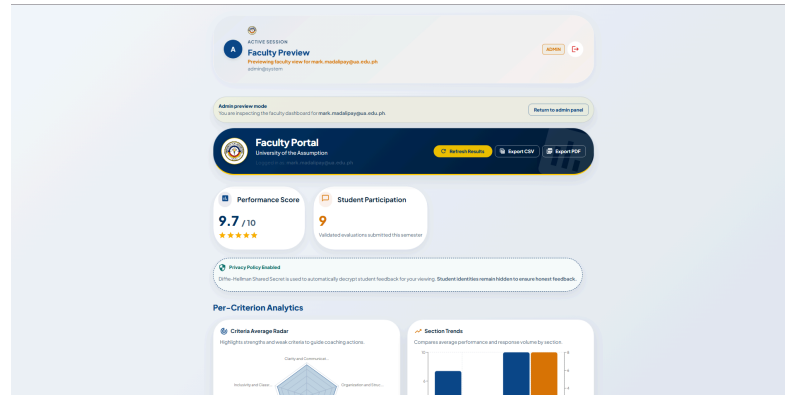


Fig. 5. Faculty Dashboard

Security Evaluation. The system's security architecture was validated through testing of the complete cryptographic workflow. Perfect Forward Secrecy was confirmed through the use of ephemeral keys, ensuring that a future compromise of the server's static key cannot expose past evaluation data. Domain-restricted OAuth 2.0 authentication was verified to correctly block access from non-institutional email addresses. All encrypted feedback stored in the database was confirmed to be unreadable without the server's private EC key, validating the Zero-Knowledge design.

Student Email	Student ID	Faculty Email	Performance	Feedback	Security
boimgaling student...	202300103	brock.vinayagius.edu.ph	5.0	tuRtCS7Wu4dAdsch4Jz-KVVCN8...	Decrypt
boimgaling student...	202300103	marbeth.iveragius.edu.ph	5.0	LzQ2tR7H7uP7u4uW7L7D7D7uW...	Decrypt
boimgaling student...	202300103	marstephen.sembrang...	5.0	W7u4uW7uW7uW7uW7uW7uW7uW...	Decrypt
antiamama student...	1231232	Antongius.edu.ph	10.0	Vu7C7uW7uW7uW7uW7uW7uW7uW...	Decrypt
antiamama student...	1231232	grace.ivaigius.edu.ph	7.8	v477uW7uW7uW7uW7uW7uW7uW...	Decrypt
gpevental student...	202100022	jerdin.mabalagius.edu...	10.0	cu477uW7uW7uW7uW7uW7uW7uW...	Decrypt
gpevental student...	202100022	mark.mabalagius.edu...	10.0	V0u477uW7uW7uW7uW7uW7uW7uW...	Decrypt
gpevental student...	202100022	mark.mabalagius.edu...	10.0	V0u477uW7uW7uW7uW7uW7uW7uW...	Decrypt
gpevental student...	202100022	mark.mabalagius.edu...	10.0	V0u477uW7uW7uW7uW7uW7uW7uW...	Decrypt
gpevental student...	202100022	mark.mabalagius.edu...	10.0	V0u477uW7uW7uW7uW7uW7uW7uW...	Decrypt

Fig. 6. Evaluation Tab

4 Conclusion and Recommendations

4.1 Conclusion

This study successfully developed and deployed CIT-EVAL, a secure, privacy-preserving online faculty evaluation system for the University of the Assumption's College of Information Technology. By integrating ECDH key exchange and AES-GCM encryption at the client side, the system effectively closes the privacy gap found in traditional evaluation platforms. The

Zero-Knowledge architecture ensures that student feedback remains confidential even from database administrators, which is expected to encourage more honest and meaningful responses.

The system's three-role structure of students, faculty, and administrators provides each stakeholder with the tools and access appropriate to their function, while maintaining strict data separation and security. The Admin Control Center and Faculty Dashboard together transform raw evaluation data into actionable insights that can support academic improvement and institutional decision-making.

4.2 Recommendations

Based on the outcomes of this study, it is recommended that the system be expanded beyond the CIT department to other colleges within the University of the Assumption following a successful and validated pilot period. Administrators should establish a secure key management policy to safeguard the server's private EC key, as its loss would render all historical encrypted feedback permanently unrecoverable. Future development should explore key recovery mechanisms or key escrow solutions to mitigate this single point of failure.

Additionally, orientation sessions should be provided to end users particularly students to familiarize them with the system's privacy features and encourage confident participation. Future research may examine the long-term impact of cryptographically secured anonymity on the quality and honesty of faculty evaluation responses, as well as explore the adoption of hardware security modules (HSM) for more robust private key storage.

References

- Diffie, W., & Hellman, M. (1976). New directions in cryptography. *IEEE Transactions on Information Theory*, 22(6), 644–654. <https://doi.org/10.1109/TIT.1976.1055638>
- Nevo, D. (2002). Evaluation in education: Current issues and practices. *Educational Evaluation and Policy Analysis*, 24(2), 124–136.
- Hassenzahl, M. (2010). Experience Design: Technology for All the Right Reasons. *Synthesis Lectures on Human-Centered Informatics*, 3(1), 1–95.
- Romero, C., & Ventura, S. (2007). Educational data mining: A review of the state of the art. *IEEE Transactions on Systems, Man, and Cybernetics, Part C*.
- Solove, D. J. (2006). A taxonomy of privacy. *University of Pennsylvania Law Review*, 154(3), 477–564.
- Boneh, D., & Shoup, V. (2023). A Graduate Course in Applied Cryptography. Retrieved from <https://crypto.stanford.edu/~dabo/cryptobook/>
- National Higher Education Research Agenda (NHERA) Form, CHED.
- Japos, G. (2019). Research Title Justification. University of the Assumption Research and Planning Office.
- Magno, C. (2013). Classroom Research Plan. University of the Assumption Research and Planning Office.
- Research and Planning Office (2013 & 2019). University of the Assumption.